



Lei Wang

RESEARCH FELLOW

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I am a hardworking, passionate, and self-disciplined researcher.

Lei Wang received his M.E. in Software Engineering from the University of Western Australia (UWA), Perth, in 2018, and his Ph.D. in Engineering and Computer Science from the Australian National University (ANU), Canberra, in 2023. He is currently a Research Fellow (Grade 2) in the School of Engineering and Built Environment — Electrical and Electronic Engineering at Griffith University. He is also a Visiting Scientist at Data61/CSIRO. He founded the Temporal Intelligence and Motion Extraction (TIME) Lab at ANU, a dynamic research team composed of master's and honours students. The lab has since become an integral part of the ARC Research Hub and the broader Griffith research community. Previously he was a Research Fellow (Level B) in the School of Computing at the ANU College of Engineering, Computing and Cybernetics and a Visiting Scientist with the Machine Learning Research Group at Data61/CSIRO (formerly NICTA). He also held Visiting Researcher positions at both the Department of Computer Science and Software Engineering at UWA and Data61/CSIRO. Since 2018, he has worked as a full-time Computer Vision Researcher at iCetana Pty Ltd., Perth, and since 2021, he has also served as a Computer Scientist at Active Intelligence Australia Pty Ltd., Perth. Lei has authored numerous first-author papers in top-tier venues, including CVPR, ICCV, ECCV, ACM MM, NeurIPS, ICLR, ICML, AAAI, TPAMI, IJCV, and TIP. His research was recognized with the Sang Uk Lee Best Student Paper Award at the Asian Conference on Computer Vision (ACCV 2022). He is currently a Guest Editor for the special issues “Trustworthy AI for Video, Web, and Social Applications” and “Motion-centric Video Processing” in the MDPI journal Electronics (Q2, h-index 110). He serves as an Area Chair for NeurIPS 2026, ACM Multimedia 2026, 2025, and 2024, ICASSP 2025, and ICPR 2024, and was recognized as an Outstanding Area Chair at ACM Multimedia 2024.

Research Interests: motion-centric | data-centric | machine learning | video understanding | one- and few-shot learning | spectral and tensor learning | optimal transportation | domain adaptation and incremental learning

Education

Doctor of Philosophy (PhD), Engineering & Computer Science , The Australian National University (ANU)	2019/07-2023/12
Master of Professional Engineering, Software Engineering , The University of Western Australia (UWA)	2016/02-2018/03
Bachelor of Engineering, Software Engineering , Yangzhou University (YZU)	2011/09-2015/06

Experience

Research Fellow (Grade 2) School of Engineering and Built Environment - Electrical and Electronic Engineering, Griffith University	2025/02-present
Visiting Scientist CSIRO's Data61, Australia	2025/03-present
Research Fellow (Level B) ANU College of Engineering, Computing and Cybernetics	2023/02-2025/02
Visiting Scientist, Postgraduate research student, Visiting Researcher CSIRO's Data61, Australia	2018/09-2024/06
Computer Scientist Active Intelligence Australia Pty Ltd, Perth, Australia	2021/07-2024/03
Machine Learning Researcher Ebenezer Technologies, Barranquilla, Colombia	2020/09-2021/01
Visiting Researcher Department of Comput. Sci. and Softw. Eng., UWA, Perth, Australia	2018/10-2019/10
Computer Vision Researcher, Computer Vision Research Intern iCetana Pty Ltd, Perth, Australia	2018/07-2020/09

Research Projects & Funding

- **Lead Chief Investigator (Lead CI)**, “LiteMotion: Learning Efficient and Scalable Video Motion Representations”, **National Computational Merit Allocation Scheme (NCMAS 2026)**, Grant ID: NCMAS-2026-39, 2,400KSU, 2026/01/01-2026/12/31
- **Chief Investigator (CI) / Delegated lead CI**, “MotionNetLite: Video Dynamics Distillation for Scalable Models”, **National Computational Merit Allocation Scheme (NCMAS 2025)**, Grant ID: NCMAS-2025-47, 1MSU and 10TB, 2025/01/01-2025/12/31
- **Lead Chief Investigator (Lead CI)**, “Efficient Video Models through Compression and Data Optimization”, **ANU Merit Allocation Scheme 2025 (ANUMAS 2025)**, 1,400KSU and 8TB, 2025/01/01-2025/12/31

- **Chief Investigator (CI) / Delegated lead CI**, “MotionNetLite: Video Dynamics Distillation for Scalable Models”, **National Computational Merit Allocation Scheme (NCMAS 2024)**, Grant ID: NCMAS-2024-67, 1MSU and 10TB, 2024/01/01-2024/12/31
- **Lead Chief Investigator (Lead CI)**, “Robust anomaly detection in human-centric videos”, The NCI **National AI Flagship Merit Allocation Scheme**, 100KSU and 5TB, 2024/01/01-2024/06/30
- **Co-Investigator (Co-I)**, “Review of Xailient’s technical pipeline of facial recognition”, **A\$ 50,815**, 2023/10/30-2023/12/22 [CI: Associate Professor Liang Zheng, Co-I: Dr Lei Wang]
- **Chief Investigator (CI) / Delegated lead CI**, “Towards building general-purpose multimodal foundation models”, **NCI Adapter Scheme Q4 2023** (HPC funding scheme), Gadi allocation 100KSU and storage allocation 25KSU (4TB), 2023/10/01-2023/12/31
- **Grant / Project Award** (Oracle Cloud credits award), “Automatic, large-scale screening of failure cases in autonomous driving”, **A\$ 48,000**, 2023/07/26-2024/07/25
- **Co-Investigator (Co-I)**, “Sharing early insights for more resilient communities”, **A\$ 71,089**, 2023/04/19-2023/10/31 [CI: Professor Lorrae Van Kerkhoff, Co-I: Mr Takuya Iwanaga, Dr Steven Lade, Dr Wendy Merritt, Dr Lei Wang, Associate Professor Liang Zheng]
- **Research sponsorship** (Active Intelligence Corp.), **Principal Investigator / Project Lead**, “Detecting anomalies in video footage (stage 3)”, **A\$ 40,013**, 2023/07/01-2024/03/01
- **Research sponsorship** (Active Intelligence Corp.), **Principal Investigator / Project Lead**, “Detecting anomalies in video footage (stage 2)”, **A\$ 108,628**, 2022/07/01-2023/06/30
- **Research sponsorship** (Active Intelligence Corp.), **Principal Investigator / Project Lead**, “Detecting anomalies in video footage”, **A\$ 135,706**, 2021/07/01-2022/06/30

Teaching

TechLauncher Examiner | School of Computing, ANU

2024/07-2024/10

- ANU TechLauncher S2 2024, co-taught courses: COMP3500, COMP4500, COMP8715
- Course Convenors: Associate Professor Charles Gretton & Associate Professor Liang Zheng
- My Student Experience of Learning and Teaching (SELT) teacher survey results: 4.11 ± 0.80 (on a 5-point scale, 413 enrollments)

TechLauncher Examiner | School of Computing, ANU

2024/02-2024/05

- ANU TechLauncher S1 2024, co-taught courses: COMP3500, COMP4500, COMP8715
- Course Convenors: Associate Professor Charles Gretton & Associate Professor Liang Zheng
- My SELT teacher survey results: 4.60 ± 0.80 (on a 5-point scale, 313 enrollments)

Teaching Assistant | Department of Comput. Sci. and Softw. Eng., UWA

2018/02-2018/06

- CITS5508 Machine Learning
- Unit Coordinator(s): Associate Professor Du Huynh & Professor Mark Reynolds

Student Supervision

- Research Interns & PhD candidates (**primary/secondary supervision**, TIME@ARC Hub&Griffith):
 - Rongchang Wang, Project title: *Appliance of vision-based action recognition in dental surgical training*, Doctoral candidate, 2025/12/08 – present
 - Zhicheng Zhang (1xWWW Companion’26 oral), Project title: *Talking-Head Generation*, Doctoral candidate@UNSW Canberra, 2025/10/10 – present
 - Syuan-Hao Li (1xWWW Companion’26), Project title: *Ultra-Fine-Grained Visual Categorization*, Bachelor Student, 2025/06/05 – present
 - Edwin Kwadwo Tenagyei (1xWWW Companion’26 oral), Project title: *Parameter Efficient Fine-Tuning*, Doctoral candidate, 2025/05/30 – present
 - Arpit Sharma (1xWWW Companion’26 oral), Project title: *Rethinking Object Detection and Tracking*, Master’s Student@Christ University, 2025/12/07 – present
- Former Research Assistants (**primary supervision**, TIME@ARC Hub&Griffith):
 - Wenxiang Diao (1xICML’26 spotlight), Project title: *Ultra-Fine-Grained Visual Categorization*, Master’s Student, 2025/05/26 – 2025/11/25
 - Ankur Singh, Project title: *Adapting TrackNetV4 for Egocentric Vision*, Bachelor Student@ANU, 2025/10/08 – 2025/11/15
 - Xi Ding (1xICLR’26, 1xAAAI’26, 1xNeurIPS’25, recipient of the **NeurIPS 2025 Scholar Award**, 2xWWW Companion’25 oral including 1xBest Paper Award), Project title: *When Features Reflect Predictions*, Master’s Student, 2024/11/13 – 2025/11/16
- Former master’s and honours students (**primary supervision**, TIME@ARC Hub&Griffith):
 - Gennie Nguyen (1xBJIT’25, 1xIEEE Access), Thesis title: *Physiological Signals for Detecting Belief and Truth in Fake News*, Master of Computing (Advanced), 2024/07/22 – 2025/05/23
 - Yifan Chen, Thesis title: *Understanding the behaviors of modern video models through signal perception*, Master of Machine Learning and Computer Vision, 2024/07/22 – 2025/05/23
 - Arjun Raj (1xICASSP’25), Thesis title: *Tracking high-speed and tiny objects in sports*, Bachelor of Advanced Computing (R&D) (Honours), 2024/07/22 – 2025/05/23
- Former master’s students (**secondary supervision**, TIME@ARC Hub&Griffith):
 - Junliang Ye (1xBMVC’25), Thesis title: *FreqSelect: Frequency-Aware Adaptive Filtering*, Master of Computing (Advanced), 2024/07/22 – 2025/05/23

- Haodong Yang, Thesis title: *SRLoRA: Subspace Recomposition in Low-Rank Adaptation via Importance-Based Fusion and Reinitialization*, Master of Machine Learning and Computer Vision, 2024/07/22 – 2025/05/23
- Meitong Liu, Thesis title: *Improving Video-to-Audio Generation via Object-Level Event Detection*, Master of Computing (Advanced), 2024/07/22 – 2025/05/23
- Former master's, honours, and internship students (**primary supervision**, TIME@ANU):
 - Dexuan Ding (1xICLR'25), Thesis title: *LEGO Fusion: Robust Video Anomaly Detection*, Bachelor of Advanced Computing (Honours), 2024/02/19 - 2024/10/25
 - Qixiang Chen (1xACML'24 oral), Thesis title: *A Closer Look at Motions*, Bachelor of Advanced Computing (Honours), 2024/02/19 - 2024/10/25
 - Huilin Chen, Thesis title: *When Spatial meets Temporal in Action Recognition*, Bachelor of Advanced Computing (Honours), 2024/02/19 - 2024/10/25
 - Liyun Zhu (1xNeurIPS'24), Thesis title: *Advancing Video Anomaly Detection*, Master of Machine Learning and Computer Vision, 2024/02/19 - 2024/10/25
 - Jushang Qiu (1xWWW Companion'25), Thesis title: *A Practical Guide to Skeletons and Taylor-transformed Skeletons in Action Recognition*, Master of Machine Learning and Computer Vision, 2024/02/19 - 2024/10/25
 - Liwen Luo, Thesis title: *Human Motion Prediction via Spatio-temporal Correspondence Learning*, Master of Machine Learning and Computer Vision, 2024/02/19 - 2024/10/25
 - Xiuyuan Yuan (1xICML'24), Project title: *Video dynamics distillation*, Summer Research Internship (ANU Summer Research Scholars Program), Bachelor of Advanced Computing (Honours), 2023/11/20 - 2024/02/02

Recognition & Honors

- **ICLR 2026 Financial Assistance Award**, International Conference on Learning Representations (ICLR), 2026/03/10
- **Best Paper Award**, The Web Conference (WWW 2025), 2025/04/29
- **Outstanding Area Chair**, ACM Multimedia 2024, 2024/10/24
- **Australian AI Awards 2024** (ANU TechLauncher), AI Innovator - Information Technology, 2024/09/19
- **iAwards 2024 ACT Merit Receipt** (ANU TechLauncher), Student & Education category, Australian Information Industry Association (AIIA), 2024/06/12
- **iAwards 2024 Finalist** (ANU TechLauncher), Business & Industry category, AIIA, 2024/05/22
- **The Sang Uk Lee Best Student Paper Award**, ACCV 2022, 2022/12/08
- **The Incentive Unit Award**, Active Intelligence Corp., 2022/09/09-2032/09/08
- **Outstanding Reviewer Award**, ECCV 2022, 2022/10/19
- **Data61 Top-up Scholarship**, Data61/CSIRO, 2019/07/01-2023/06/30
- **Data61 PhD Scholarship**, Data61/CSIRO, 2019/07/01-2023/06/30
- **ANU HDR Fee Remission Merit Scholarship**, ANU, 2019/07/01-2023/06/30
- **Outstanding Graduate Award**, YZU, 2015/06
- **Jingwen Zhu Scholarship**, YZU, 2015/03
- **Principal's Scholarship (First Division)**, YZU, 2014/12
- **The Second Prize of Lanqiao Cup Competition (C/C++ Program Design)**, National Software and Information Technology Professional Talent Competition, 2014/04
- **Principal's Scholarship (Second Division)**, YZU, 2013/12

Professional Service

- Area Chair: NeurIPS 2026, ACM Multimedia 2026, ACM Multimedia 2025, ICASSP 2025, ACM Multimedia 2024, ICPR 2024
- Guest Editor, Electronics (MDPI) – Special Issues: “Trustworthy AI for Video, Web, and Social Applications” and “Motion-centric Video Processing” [Q2, h-index 110]
- Reviewer:
 - Journals: TPAMI, IJCV, TIP, ACM CSUR, TCSVT, TII, TMM, TIFS, TKDE, PR, BDMA, CVIU, Neurocomputing, ACM TOMM, IEEE Signal Processing Letters, IEEE Access, CAAI Transactions on Intelligence Technology, Journal of Medical Internet Research (JMIR), NCAA, IMAVIS, JVCI, SIVP, TVCJ, Cognitive Robotics, SN Computer Science, Algorithms, Multimedia Systems, Machine Learning and Knowledge Extraction (MAKE), Nuclear Engineering and Technology, International Journal of Clinical and Health Psychology (IJCHP), Sensors, Electronics, Neural Networks, Journal of Imaging, Future Internet, Information, Healthcare, IET Computer Vision, Electronics Letters, AI Communications, Global Health Economics and Sustainability, PeerJ Computer Science, Computers, AI Computer Science and Robotics Technology (ACRT), Artificial Intelligence and Emerging Technologies (AIET)
 - Conferences: NeurIPS 2022-2025, ICLR 2023-2026, AAAI 2022-2026, ICML 2022-2026, AISTATS 2025-2026, CVPR 2022-2026, ICCV 2023-2025, ECCV 2022-2026, WACV 2024-2026, BMVC 2020-2025, ACCV 2024-2026, ICASSP 2025-2026, ICME 2025-2026, IJCNN 2025, CSCW 2026, IEEE VIS 2025, IEEE CAI 2023, MIND 2025
 - Workshops: Computer Vision for Metaverse Workshop (CV4Metaverse at ECCV 2024, CVPR 2025), The AI City Challenge Workshop (CVPR 2023-2024, ICCV 2025), Vision Datasets Understanding (CVPR 2022-2024, ICCV 2025), Deep Vision Work-

shop (CVPR 2020)

- Thesis Examiner for Honors and Master's Theses: ANU College of Engineering, Computing & Cybernetics, S1 & S2 2023, S1 & S2 2024, S1 2025 (external examiner)
- Workshop organizer: (i) "TIME 2025: 1st International Workshop on Transformative Insights in Multi-faceted Evaluation" at The Web Conference (WWW 2025), (ii) "TIME 2026: 2nd International Workshop on Transformative Insights in Multi-faceted Evaluation" at The Web Conference (WWW 2026)
- Conference service (session chair): AAAI 2026 (Singapore EXPO, Singapore): (i) AAAI/ACM SIGAI Doctoral Dissertation Talk session, and (ii) ML: Graph-Based Machine Learning 4 session

Invited Talk

- Feature Hallucination for Self-supervised Action Recognition | AAAI-26 Journal Track Program, Singapore EXPO, Singapore | 2026/1/23
- Beyond Pixels, Toward Understanding | ARC Industrial Transformation Research Hub for Driving Farming Productivity and Disease Prevention, Griffith University (Nathan Campus) | 2025/10/14
- Action Recognition: Past, Present and Future | Department of Computer Science, Harbin Institute of Technology | 2023/08/12

Publications

Journal impact factors follow Clarivate JCR 2023; conference rankings follow CORE 2023.

* denotes co-first authors (equal contribution); † denotes corresponding author.

- Journals

- [j9] – Edwin Kwadwo Tenagyei*, **Lei Wang***, Yongsheng Gao, Jun Zhou, Piotr Koniusz. "Behavior-Aligned Fine-Tuning without Extra Parameters." *Pattern Recognition*, 2026. **IF: 7.6 Under revision**
- [j8] – Yang Liu, Jiyao Yang, Madhawa Perera, Pan Ji, Dongwoo Kim, Min Xu, Tianyang Wang, Saeed Anwar, Tom Gedeon, **Lei Wang**†, Zhenyue Qin. "Representation-Centric Survey of Skeletal Action Recognition and the New Benchmark." *Pattern Recognition*, 2026. **IF: 7.6 Under revision**
- [j7] – Gennie Nguyen, **Lei Wang**, Yangxueqing Jiang, Tom Gedeon. "Detecting Fake News Belief via Skin and Blood Flow Signals." *IEEE Access*, 2026.
- [j6] – Gennie Nguyen, **Lei Wang**, Yangxueqing Jiang, Tom Gedeon. "Truth and Trust: Fake News Detection via Biosignals." *International Journal of Information Technology (BJIT)*, 2025.
- [j5] – **Lei Wang**† and Piotr Koniusz. "Feature Hallucination for Self-supervised Action Recognition." *International Journal of Computer Vision*. 2025. **IF: 11.6**
- [j4] – **Lei Wang**†, Jun Liu, Liang Zheng, Tom Gedeon, and Piotr Koniusz. "Meet JEANIE: a Similarity Measure for 3D Skeleton Sequences via Temporal-Viewpoint Alignment." *International Journal of Computer Vision*. 2024. **IF: 11.6**
- [j3] – Zhenyue Qin, Yang Liu, Pan Ji, Dongwoo Kim, **Lei Wang**, R.I. (Bob) McKay, Saeed Anwar, and Tom Gedeon. "Fusing Higher-Order Features in Graph Neural Networks for Skeleton-Based Action Recognition." *IEEE Transactions on Neural Networks and Learning Systems* (2022): 4783-4797. **IF: 10.2**
- [j2] – Piotr Koniusz, **Lei Wang**, and Anoop Cherian. "Tensor Representations for Action Recognition." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44.2 (2021): 648-665. **IF: 20.8**
- [j1] – **Lei Wang**†, Du Q. Huynh, and Piotr Koniusz. "A Comparative Review of Recent Kinect-based Action Recognition Algorithms." *IEEE Transactions on Image Processing*, 29 (2019): 15-28. **IF: 10.8**

- Conferences

- [c24] – Haozhe Jia, Pengyu Yin, Wenshuo Chen, Shaofeng Liang, **Lei Wang**, Bowen Tian, Xiucheng Wang, Jia Nanqian, Yutao Yue. "Learning to Think in Physics: Breaking Shortcut Learning in Scientific Diffusion via Representation Alignment." *International Conference on Machine Learning (ICML)*. 2026. **A***
- [c23] – **Lei Wang**, Wenxiang Diao, Andrew Busch, Jun Zhou, Yongsheng Gao. "Privacy-Aware Video Anomaly Detection through Orthogonal Subspace Projection." *International Conference on Machine Learning (ICML)*. 2026. **A*, spotlight**
- [c22] – **Lei Wang**, Xi Ding, Yongsheng Gao, Piotr Koniusz. "Subspace Kernel Learning on Tensor Sequences." *International Conference on Learning Representations (ICLR)*, 2026. **A***
- [c21] – Xi Ding*, **Lei Wang***, Piotr Koniusz, Yongsheng Gao. "Learning Time in Static Classifiers." *AAAI Conference on Artificial Intelligence (AAAI)*, 2026. **A***
- [c20] – Xi Ding*, **Lei Wang***, Piotr Koniusz, Yongsheng Gao. "Graph Your Own Prompt." *Advances in Neural Information Processing Systems (NeurIPS)*. 2025. **A***
- [c19] – Jiahao Ma, **Lei Wang**, Miaomiao Liu, David Ahmedt-Aristizabal, Chuong Nguyen. "Puzzles: Unbounded Video-Depth Augmentation for Scalable End-to-End 3D Reconstruction." *Advances in Neural Information Processing Systems (NeurIPS)*. 2025. **A***
- [c18] – Haozhe Jia*, Wenshuo Chen*, Zhihui Huang*, **Lei Wang**, Hongru Xiao, Nanqian Jia, Keming Wu, Songning Lai, Bowen Tian, Yutao Yue. "Physics-Informed Representation Alignment for Sparse Radio-Map Reconstruction." *ACM Multimedia*,

2025. **A***, **oral**

[c17] – Junliang Ye, **Lei Wang**[†], Md Zakir Hossain. “FreqSelect: Frequency-Aware fMRI-to-Image Reconstruction.” *British Machine Vision Conference (BMVC)*, 2025. **A**

[c16] – Wenshuo Chen*, Kuimou Yu*, Haozhe Jia*, Kaishen Yuan, Zexu Huang, Bowen Tian, Songning Lai, Hongru Xiao, Erhang Zhang, **Lei Wang**, Yutao Yue. “ANT: Adaptive Neural Temporal-Aware Text-to-Motion Model.” *ACM Multimedia*, 2025. **A***

[c15] – Dexuan Ding, **Lei Wang**[†], Liyun Zhu, Tom Gedeon, Piotr Koniusz. “Learnable Expansion of Graph Operators for Multi-Modal Feature Fusion.” *International Conference on Learning Representations (ICLR)*, 2025. **A***

[c14] – Arjun Raj, **Lei Wang**[†], Tom Gedeon. “TrackNetV4: Enhancing Fast Sports Object Tracking with Motion Attention Maps.” *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2025. **B**

[c13] – Liyun Zhu, **Lei Wang**[†], Arjun Raj, Tom Gedeon, and Chen Chen. “Advancing Video Anomaly Detection: A Concise Review and a New Dataset.” *Advances in Neural Information Processing Systems (NeurIPS)*. 2024. **A***

[c12] – Qixiang Chen, **Lei Wang**[†], Piotr Koniusz and Tom Gedeon. “Motion meets Attention: Video Motion Prompts.” *Asian Conference on Machine Learning (ACML)*. 2024. **Long presentation (Conf. track: 20.11% accept., 6.52% long pres.)**

[c11] – Wenshuo Chen*, Hongru Xiao*, Erhang Zhang*, Lijie Hu, **Lei Wang**, Mengyuan Liu, Chen Chen. “SATO: Stable Text-to-Motion Framework.” *ACM Multimedia*, 2024. **A***

[c10] – **Lei Wang**[†], Piotr Koniusz, Tom Gedeon, and Liang Zheng. “Adaptive Multi-head Contrastive Learning.” *European Conference on Computer Vision*. 2024. **A***

[c9] – **Lei Wang**[†], Xiuyuan Yuan, Tom Gedeon, and Liang Zheng. “Taylor Videos for Action Recognition.” *International Conference on Machine Learning (ICML)*. 2024. **A***

[c8] – **Lei Wang**, Ke Sun, and Piotr Koniusz. “High-order Tensor Pooling with Attention for Action Recognition.” *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023. **B, oral**

[c7] – **Lei Wang** and Piotr Koniusz. “Flow Dynamics Correction for Action Recognition.” *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2023. **B**

[c6] – **Lei Wang** and Piotr Koniusz. “3Mformer: Multi-order Multi-mode Transformer for Skeletal Action Recognition.” *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2023. **A***

[c5] – **Lei Wang** and Piotr Koniusz. “Uncertainty-DTW for Time Series and Sequences.” *European Conference on Computer Vision*. Cham: Springer Nature Switzerland, 2022. **A*, oral**

[c4] – **Lei Wang** and Piotr Koniusz. “Temporal-Viewpoint Transportation Plan for Skeletal Few-shot Action Recognition.” *Proceedings of the Asian Conference on Computer Vision*. 2022. **B, oral, Best Student Paper Award**

[c3] – **Lei Wang** and Piotr Koniusz. “Self-supervising Action Recognition by Statistical Moment and Subspace Descriptors.” *Proceedings of the 29th ACM international conference on multimedia*. 2021. **A***

[c2] – **Lei Wang**, Piotr Koniusz, and Du Q. Huynh. “Hallucinating IDT Descriptors and I3D Optical Flow Features for Action Recognition with CNNs.” *2019 IEEE/CVF International Conference on Computer Vision*, IEEE Computer Society, 2019. **A***

[c1] – **Lei Wang**, Du Q. Huynh, and Moussa Reda Mansour. “Loss Switching Fusion with Similarity Search for Video Classification.” *2019 IEEE International Conference on Image Processing (ICIP)*, IEEE, 2019. **B, 1 AU patent**

- Workshops

[w11] – Arpit Sharma, **Lei Wang**[†], Yongsheng Gao. “Rethinking Object Detection and Tracking.” *Companion Proceedings of the ACM Web Conference*, 2026. **Oral**

[w10] – Edwin Kwadwo Tenagyei, **Lei Wang**[†], Yongsheng Gao. “Beyond Fine-Tuning: The Present and Future of Parameter-Efficient Fine-Tuning in Vision Transformers.” *Companion Proceedings of the ACM Web Conference*, 2026. **Oral**

[w9] – Haosen Li, Wenshuo Chen, Shaofeng Liang, **Lei Wang**, Haozhe Jia, Yutao Yue. “Guided Path Sampling: Steering Diffusion Models Back on Track with Principled Path Guidance.” *Companion Proceedings of the ACM Web Conference*, 2026. **Oral**

[w8] – Peilin Zheng, Xiucheng Wang, Wenshuo Chen, Ruijin Sun, **Lei Wang**, Yutao Yue, Nan Cheng. “RadioAR: Autoregressive Modeling for Accurate Radio Map Estimation.” *Companion Proceedings of the ACM Web Conference*, 2026. **Oral**

[w7] – Syuan-Hao Li, Yi-Jui Huang, **Lei Wang**, Yongsheng Gao. “Fine-Grained Visual Understanding for Multimodal and Trustworthy AI.” *Companion Proceedings of the ACM Web Conference*, 2026.

[w6] – Wenshuo Chen, Haozhe Jia, Kuimou Yu, Songning Lai, **Lei Wang**, Yutao Yue. “Towards Better Evaluation Metrics for Text-to-Motion Generation.” *Companion Proceedings of the ACM Web Conference*, 2026. **Oral**

[w5] – Zhicheng Zhang*, **Lei Wang***, Yu Zhang, Yongsheng Gao. “Talking-Head Generation in Practice.” *Companion Proceedings of the ACM Web Conference*, 2026. **Oral**

[w0] – **Lei Wang**[†], Md Zakir Hossain, Tom Gedeon, Zicheng Pan, Syed Islam, Md Rafiqul Islam, Yasmeen George, Shreya Ghosh. “TIME 2026: 2nd International Workshop on Transformative Insights in Multi-faceted Evaluation.” *Companion Proceedings of the ACM Web Conference*, 2026.

[w4] – Sen Fang, Chen Chen, **Lei Wang**, Ce Zheng, Chunyu Sui, Yapeng Tian. “SignLLM: Sign Language Production Large Language Models.” *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*, 2025.

[w3] – Xi Ding, **Lei Wang**[†]. “The Journey of Action Recognition.” *Companion Proceedings of the ACM Web Conference*, 2025. **Oral, Best Paper Award**

[w2] – Xi Ding, **Lei Wang**[†]. “Do Language Models Understand Time?” *Companion Proceedings of the ACM Web Conference*, 2025. **Oral**

[w1] – Jushang Qiu, **Lei Wang**[†]. “Evolving Skeletons: Motion Dynamics in Action Recognition.” *Companion Proceedings of the ACM Web Conference*, 2025.

[w0] – **Lei Wang**[†], Md Zakir Hossain, Syed Islam, Tom Gedeon, Sharifa Alghowinem, Isabella Yu, Serena Bono, Xuanying Zhu, Gennie Nguyen, Nur Al Hasan Haldar, Seyed Mohammad Jafar Jalali, Md Abdur Razzaque, Imran Razzak, Md Rafiqul Islam, Shahadat Uddin, Naeem Janjua, Aneesh Krishna, Manzur Ashraf. “TIME 2025: 1st International Workshop on Transformative Insights in Multi-faceted Evaluation.” *Companion Proceedings of the ACM Web Conference*, 2025.

- Patents

[p3] – **Lei Wang**. “System and Method of Detecting Anomalies from Mass Data.” (US provisional, SN 63/326,525)

[p2] – **Lei Wang** and Graeme Woods. “Method and System for Classifying Video Data.” (au 2019903775, provisional patent filed 07/10/2019)

[p1] – **Lei Wang**, Moussa Reda Mansour, and Graeme Woods. “System and Method of Video Data Retrieval.” (au 2019900316, provisional patent filed 01/02/2019)

- Theses

[t2] – **Lei Wang**. “Robust Human Action Modelling.” *PhD thesis*, The Australian National University, Canberra, ACT, Australia, Nov. 2023.

[t1] – **Lei Wang**. “Analysis and evaluation of Kinect-based action recognition algorithms.” *Master’s thesis*, The University of Western Australia, Perth, WA, Australia, Nov. 2017.